

Guarding the Recommendation Pipeline

SVD/ALS Factorization with Fake Review Filtering

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Abstract

Modern e-commerce platforms frequently struggle to deliver reliable, high-fidelity product suggestions due to the extreme sparsity of user preference matrices, where the average consumer interacts with only a minute fraction of the global inventory. Under the core theme of **Recommender Systems via Matrix Factorization**, this project addresses the limitation by building a collaborative filtering engine engineered to uncover hidden patterns in user preferences by decomposing highly sparse preference matrices. The system operates through an item-centric workflow: given a specific product query as input, it uncovers and extracts a curated list of top-rated, highly relevant alternatives. To achieve this, the sparse preference matrix $R \in \mathbb{R}^{m \times n}$ is subjected to low-rank decomposition using **Truncated Singular Value Decomposition (SVD)** and a regularized latent-factor model optimized via **Alternating Least Squares (ALS)**. By mapping both users and items into a shared latent factor space, the engine systematically minimizes iterative matrix approximation errors to capture complex, non-obvious relationships between products, enabling the accurate prediction of missing ratings and robust similarity computations.

To ensure the absolute integrity and trustworthiness of these suggestions, the collaborative filtering pipeline is integrated with an advanced **Natural Language Processing (NLP) and Machine Learning** guardrail dedicated to detecting AI-generated fake reviews and deceptive spam. This security layer analyzes review text structures, sentiment-rating anomalies, and bot-like behavioral footprints to filter out fraudulent feedback and rating inflation *before* the data is processed by the factorization engine. By neutralizing malicious data distortions at the input stage, the matrix decomposition reflects true, organic user preference patterns, ensuring the system recommends only genuinely high-performing items. The performance and scalability of the architecture are rigorously evaluated using **Root Mean Squared Error (RMSE)** and **Mean Absolute Error (MAE)** metrics across varying latent dimensions and regularization penalties. Implemented in Python using optimized sparse-matrix representations, this project successfully bridges classical numerical linear algebra with supervised learning to deliver a robust, scalable, and fraud-resistant personalization engine for large-scale digital marketplaces.